Q1.

If $g(x) = 4x^2 - 2x + 1$, $f(1) = -1$ and $g(f(x)) = x^2 - 7x + 13$, then the minimum value of $f(g(x))$ is equal to:

(1) $-\dfrac{9}{8}$ (2) $\dfrac{3}{4}$ (3) $-\dfrac{9}{4}$ (4) -1

Q2.

Let a circle C_1 pass through the origin and have its center on the line $y = 2x$. If the line $x + 2y - 10 = 0$ is a tangent to the circle C_1, let the point of contact be P. Let C_2 be another circle described on the line segment OP as its diameter, where O is the origin. The length of the intercept made by the circle C_2 on the x-axis is equal to:

(1) 4 (2) 5 (3) 2 (4) 10

Q3.

The value of $\sum_{k=1}^{\infty}(-1)^{k+1}\left(\dfrac{k^2+1}{(k+1)!}\right)$ is:

(1) $\dfrac{4-e}{e}$ (2) $\dfrac{4+e}{e}$ (3) $\dfrac{2-e}{e}$ (4) $\dfrac{e-4}{e}$

Q4.

If $\tan A$ and $\tan B$ are the roots of the quadratic equation $x^2 - px + q = 0$, where $q \neq 1$, then $\sin^2(A+B)$ is equal to:

(1) $\dfrac{p^2}{p^2+(1-q)^2}$ (2) $\dfrac{q^2}{p^2+(1-q)^2}$
(3) $\dfrac{p^2}{q^2+(1-p)^2}$ (4) $\dfrac{p^2}{(p+q)^2}$

Q5.

Let $x - y = 0$ and $y = 0$ be the internal angle bisectors of angles B and C of $\triangle ABC$. If the side BC lies on the line $3x - y = 5$ and the vertex A is (α, β), then the value of $\alpha^2 + \beta^2$ is:

(1) 3 (2) 4 (3) 5 (4) 6

Q6.

Let z be a complex number such that $\arg(z-2)=\dfrac{\pi}{4}$ and $|z-4|=\sqrt{2}$. Then the value of $z^3-4z^2-2z+35$ is equal to

(1) 15 (2) 20 (3) 25 (4) 30

Q7.

A bag contains 6 balls. The colour of each ball is independently determined by tossing a fair coin, making it either red or black. If three balls are drawn at random without replacement from the bag and all are found to be black, then the probability that the bag originally contained exactly 1 red ball is:

(1) $2/7$ (2) $3/8$ (3) $1/4$ (4) $1/8$

Q8.

Let $S=\{x^3+ax^2+bx+c : a,b,c\in\mathbb{N}$ and $a,b,c\le 12\}$ be a set of polynomials. Then the number of polynomials in S, which have two roots whose sum is zero, is equal to:

(1) 42 (2) 35 (3) 28 (4) 30

Q9.

Let $S=\{1,2,3,4,5,6,7\}$. Let x be the total number of permutations of the set S such that exactly four elements remain in their original positions. Let y be the total number of permutations of the set S such that exactly three elements remain in their original positions. Then, which of the following relations is correct?

(1) $9x=2y$
(2) $2x=9y$
(3) $7x=3y$
(4) $3x=7y$

Q10.

Let α,β be the roots of $x^2-(a-2)x-(a+1)=0$. If $\alpha^2+\beta^2$ attains its minimum value at $a=\lambda$, then $\alpha^3+\beta^3$ evaluated at $a=\lambda$ is

(1) -7 (2) 7 (3) -9 (4) 9

Q11.

Let $I=\int_0^{3\pi} x\,(|\sin x-\cos x|+|\sin x+\cos x|)dx$. If the value of the integral I is expressed as $k\pi\sqrt{2}$, then the value of k is equal to:

(1) 9 (2) 12 (3) 18 (4) 36

Q12.

The value of

$$\lim_{x\to 0} \frac{\int_0^x t\left(e^{2\sec t}+e^{4\sec t}+\cdots+e^{20\sec t}\right)dt}{\ln(1+e^2x^2)}$$

is equal to

(1) $\dfrac{e^{10}-1}{2e^2(e^2-1)}$

(2) $\dfrac{e^{20}-1}{2e^2(e^2-1)}$

(3) $\dfrac{e^{20}-1}{2(e^2-1)}$

(4) $\dfrac{e^{10}-1}{2(e^2-1)}$

Q13.

The area of the region $R=\{(x,y): xy\le 64,\ 4\le x\le y^2,\ y\ge 0\}$ is

(1) $\dfrac{8}{3}(48\log_e(2)-15)$

(2) $\dfrac{16}{3}(24\log_e(2)-7)$

(3) $\dfrac{32}{3}(12\log_e(2)-7)$

(4) $\dfrac{16}{3}(20\log_e(2)-9)$

Q14.

The mean and variance of 8 observations are 9 and 21, respectively. If 6 of these observations are $2, 4, 10, 12, 14, 16$, then the variance of the absolute deviations of all 8 observations from their mean is:

(1) 4 (2) 5 (3) 16 (4) 21

Q15.

Let $a_1, a_2, \ldots$ be an A.P. with a non-zero common difference d. If the terms a_2, a_5, and a_{14} form a geometric progression, and the sum of the first 10 terms of the A.P. is 200, then a_{20} is equal to

(1) 42 (2) 74 (3) 78 (4) 82

Q16.

If $\int e^{\sec x}(\cos x+\tan^2 x)\,dx = f(x)+C$, where C is the constant of integration, then $f\left(\frac{\pi}{3}\right)-f(0)$ is equal to:

(1) $\frac{1}{2}e^2$ (2) $\frac{\sqrt{3}}{2}e^2$ (3) $\sqrt{3}e^2$ (4) e^2-1

Q17.

Let $\vec{p}, \vec{q}, \vec{r}$ be unit vectors satisfying $|\vec{p}+\vec{q}+\vec{r}|=1$ and $\vec{p}\cdot\vec{q}=0$. The value of $|\vec{p}\times\vec{r}+\vec{q}\times\vec{r}|^2$ is equal to:

(1) 1 (2) 2 (3) 3 (4) $\frac{1}{2}$

Q18.

Let A be a 2×2 matrix with real entries such that the trace of A is 3 and $|A|=2$. If $B=A^3-2A^2+A-I$, then the determinant of B is equal to:

(1) -1 (2) 1 (3) 5 (4) -5

Q19.

Let a curve $x=x(y)$ be the solution of the differential equation $\frac{dy}{dx}=\frac{y}{x+y^2\cos y}$, for $y>0$. If $x(\pi)=\pi$, then the area bounded by the curve, the y-axis, and the lines $y=0$ and $y=\frac{\pi}{2}$ is equal to:

(1) $\frac{\pi^2+8}{8}$ (2) $\frac{\pi^2+4}{4}$
(3) $\frac{\pi^2-8}{8}$ (4) $\frac{\pi^2}{8}$

Q20.

Let the direction cosines l, m, n of two lines L_1 and L_2 satisfy the relations $l+m+n=0$ and $2ln+mn=0$. If $P(1,-1,2)$ and $Q(2,1,2)$ are two points, then the sum of the squares of the projections of the line segment PQ on the lines L_1 and L_2 is equal to:

(1) 6 (2) 2 (3) 3 (4) 5

Q21.

Let $\{a_n\}$ be a geometric progression of positive real numbers such that $a_1=2$ and $a_{n+1}-2a_n=0$ for all $n\geq 1$. If a new sequence $\{b_n\}$ is defined by $b_n=\log_2(a_n)$, then the value of the sum $\sum_{n=1}^{10} b_n$ is equal to ____.

Q22.

If $k=\tan\left(\cot^{-1}(1)+\cot^{-1}(3)+\cot^{-1}(7)\right)$, then the number of real solutions of the equation $\cos^{-1}(x)+\cos^{-1}(2x)=\frac{k\pi}{9}$ is ___.

Q23.

Let $\vec{PQ} = \hat{i} + 2\hat{j} + 2\hat{k}$ and $\vec{PR} = 2\hat{i} + 3\hat{j} - \hat{k}$ be two adjacent sides of a triangle PQR. Let $\vec{v}$ be a vector in the plane of the triangle such that $\vec{v}$ is perpendicular to $\vec{PQ}$. If the dot product of $\vec{v}$ with $\vec{PR}$ is 30, then the value of $|\vec{v}|^2$ is ___.

Q24.

For some $\theta \in \left(0, \frac{\pi}{2}\right)$, let e_1 and d_1 be the eccentricity and the distance between the foci of the hyperbola $x^2 - y^2 \cot^2 \theta = 2$, respectively, and let e_2 and d_2 be the eccentricity and the distance between the foci of the ellipse $x^2 \sec^2 \theta + y^2 = 8$, respectively. If $d_1^2 = d_2^2 \sec^2 \theta - 16$, then $d_1 d_2 (e_1^2 - e_2^2) \sin^2 \theta$ is equal to ____.

Q25.

If $I_k = \int_0^{\frac{\pi}{4}} \tan^k x \, dx$, then the value of $\sum_{k=1}^{15} k(k+1)(I_k + I_{k+2})$ is ____.

Q26.

The electric current in a circuit is given by the expression $i = i_0 \sqrt{t/T}$ for the period $t = 0$ to $t = T$. The total heat produced in a resistor R during this time interval is

(1) $\frac{1}{2} i_0^2 RT$ (2) $i_0^2 RT$ (3) $\frac{1}{3} i_0^2 RT$ (4) $\frac{1}{\sqrt{2}} i_0^2 RT$

Q27.

Two identical bulbs, each of resistance R, are connected in series across a battery of internal resistance r. The total power dissipated in the bulbs is P. When these bulbs are connected in parallel across the same battery, the total power dissipated in them becomes $2.25P$. The ratio R/r is:

(1) 2 (2) 4 (3) 6 (4) 8

Q28.

4 kg of ice at -20°C is added to 20 kg of water at 10°C in a perfectly insulated container. Once the system reaches thermal equilibrium, what is the ratio of the mass of water to the mass of ice in the final mixture?
(specific heat of ice = 2100 J/kg·°C, specific heat of water = 4200 J/kg·°C, latent heat of fusion of ice $= 3.36 \times 10^5$ J/kg)

(1) 11 (2) 10 (3) 15 (4) 1

Q29.

Three identical thin rods, each of mass $M = 0.6$ kg and length $L = 30$ cm, are joined to form a rigid equilateral triangle. The system is rotated about an axis passing through one of its vertices and perpendicular to the plane of the triangle. If a constant torque of 0.405 N-m is applied, the angular acceleration of the system is

(1) 2 rad/s^2 (2) 3 rad/s^2 (3) 5 rad/s^2 (4) 8 rad/s^2

Q30.

An elevator starts from rest and rises with a constant acceleration of 2 m/s^2. Coins are dropped from it at regular time intervals. The first coin is dropped at $t = 0$ s, and the fourth coin is dropped at $t = 3$ s. The distance between the second and third coins at $t = 3$ s is ____ m. (Assume the coins have not hit the floor and $g = 10$ m/s^2)

(1) 12 (2) 15 (3) 18 (4) 24

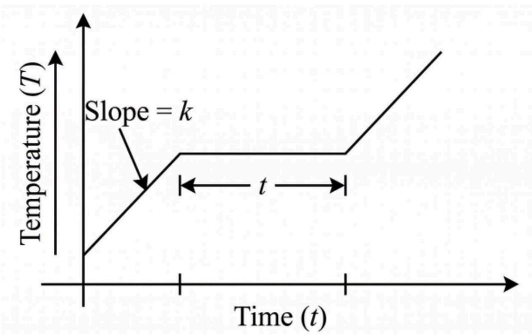
Q31.

The magnetic field of an electromagnetic wave travelling through a non-magnetic medium ($\mu_r = 1$) is given by $B(y, t) = 10^{-7} \sin(10^3 y + 2 \times 10^{11} t)\hat{k}$ (all values in SI units). The relative permittivity (ϵ_r) of the medium is:

(1) 1.5 (2) 2.25 (3) 4.0 (4) 9.0

Q32.

A solid is heated at a constant power. Its temperature-time graph shows a solid-phase slope k, and a melting plateau of duration t. The ratio of its latent heat of fusion to solid specific heat (L/s) is:



(1) k/t (2) t/k (3) kt (4) $1/(kt)$

Q33.

Three long straight parallel wires are passing through the vertices of an equilateral triangle of side 10 cm, such that the wires are perpendicular to the plane of the triangle. Wires at vertices A and B each carry a current of 10 A directed into the plane, while the wire at vertex C carries 10 A directed out of the plane. The magnitude of the force experienced by a 25 cm long section of the wire at vertex C is

(1) 5×10^{-5} N
(2) $5\sqrt{3} \times 10^{-5}$ N
(3) 2.5×10^{-4} N
(4) $2.5\sqrt{3} \times 10^{-4}$ N

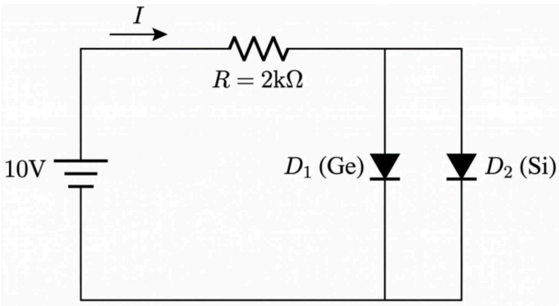
Q34.

A biconvex lens of refractive index 1.5 acts as a concave mirror of focal length magnitude F_1 when its back surface is silvered. When its front surface is silvered instead, it acts as a concave mirror of focal length magnitude F_2. If $F_1 = 2F_2$, then the ratio of the radius of curvature of the front surface to that of the back surface of the lens is ____.

(1) 5 : 1
(2) 1 : 5
(3) 2 : 1
(4) 1 : 2

Q35.

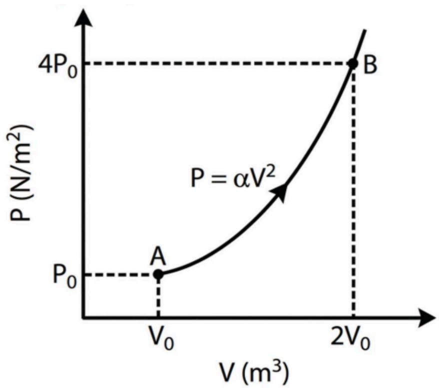
In the circuit shown below, D_1 is a Germanium (Ge) diode and D_2 is a Silicon (Si) diode. The forward voltage drops for Ge and Si are 0.3 V and 0.7 V, respectively. The current I drawn from the 10 V source is:



(1) 4.65 mA
(2) 4.85 mA
(3) 2.32 mA
(4) 5.00 mA

Q36.

In the following $P - V$ diagram, an ideal monoatomic gas is taken from state A to state B. The equation of state along the curved path is given by $P = \alpha V^2$, where α is a positive constant. The total heat supplied to the gas in this process is:



(1) $\dfrac{21}{2} P_0 V_0$ (2) $13 P_0 V_0$

(3) $\dfrac{77}{6} P_0 V_0$ (4) $\dfrac{7}{3} P_0 V_0$

Q37.

A stationary radioactive nucleus undergoes spontaneous fission into two fragments. If the ratio of the surface areas of the newly formed daughter nuclei is 1 : 4, what is the ratio of their respective kinetic energies immediately after the fission?

(1) 1 : 8 (2) 8 : 1 (3) 1 : 2 (4) 4 : 1

Q38.

Two cells of EMF 6 V and 12 V have the same internal resistance r, are connected to an external resistor of resistance $R = 8\,\Omega$. The current passing through the external resistor R is found to be the same whether the cells are connected in series (assisting each other) or in parallel (with like terminals connected together). The value of the internal resistance r is

(1) $2\,\Omega$ (2) $4\,\Omega$ (3) $8\,\Omega$ (4) $10\,\Omega$

Q39.

A block of mass 2 kg is on an inclined plane at an angle of 37° to the horizontal. The coefficient of friction is 0.5. A horizontal force F is applied to the block. The value of F such that the block moves up the incline with constant velocity is: (Take $g = 10$ m/s^2, $\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$)

(1) 20 N (2) 30 N (3) 40 N (4) 50 N

Q40.

Given below are two statements:

Statement-I: A plane wavefront incident on a thin convex lens emerges as a converging spherical wavefront, whereas passing through a thin concave lens it emerges as a diverging spherical wavefront.

Statement-II: If a standard Young's double-slit experimental setup is completely immersed in a liquid of refractive index $\mu > 1$, the geometrical path difference between the waves at any point on the screen remains unchanged, but the fringe width increases.

In the light of the above statements, choose the correct answer from the options given below.

(1) Both Statement-I and Statement-II are false.
(2) Both Statement-I and Statement-II are true.
(3) Statement-I is true but Statement-II is false.
(4) Statement-I is false but Statement-II is true.

Q41.

A uniform heavy rope of mass M and length L hangs vertically from a rigid support, elongating by Δl_1 under its own weight. When a point block of mass $2M$ is attached to its free lower end, the total elongation of the rope becomes Δl_2. The ratio $\Delta l_1 : \Delta l_2$ is $1 : x$. The value of x is ____.

(1) 3 (2) 4 (3) 5 (4) 6

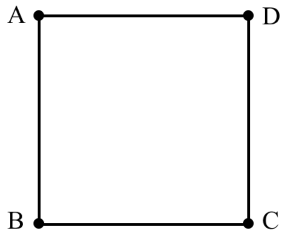
Q42.

In a Vernier caliper, 20 Vernier scale divisions coincide with 19 main scale divisions (1 MSD $=$ 1 mm). When both jaws touch, the zero of the Vernier scale is to the left of the main scale zero, and the 14th Vernier division coincides with a main scale mark. While measuring a cylinder's length, the main scale reads 32 mm and the 12th Vernier division coincides. The true length of the cylinder is ____ mm.

(1) 32.30 (2) 32.60 (3) 32.90 (4) 33.30

Q43.

Three point charges $q_A = +1$ nC, $q_B = -2$ nC, and $q_C = +1$ nC are fixed at three corners of a square of side 3 cm. The work done in bringing a charge $q_D = +4$ nC from infinity to the fourth corner of the square is:



(1) $1.2(2 - \sqrt{2})\ \mu$J
(2) $1.2(2 + \sqrt{2})\ \mu$J
(3) $0.6(4 - \sqrt{2})\ \mu$J
(4) Zero

Q44.

Two identical coaxial circular loops of radius R carry the same current in the same direction. Their centers are separated by a distance $\sqrt{3}R$. If the magnetic field produced by a single loop at its own center is 80 μT, the net magnetic field at the center of either loop is ____ μT.

(1) 90
(2) 70
(3) 10
(4) 160

Q45.

A particle of mass m falls from rest through a resistive medium having resistive force $F = -kv$, where v is the velocity and k is a constant. It eventually attains a terminal velocity v_0. What is the rate of work done by gravity on the particle at the exact instant its downward acceleration is $g/2$?

(1) mgv_0
(2) $\frac{1}{4}mgv_0$
(3) $\frac{1}{2}mgv_0$
(4) Zero

Q46.

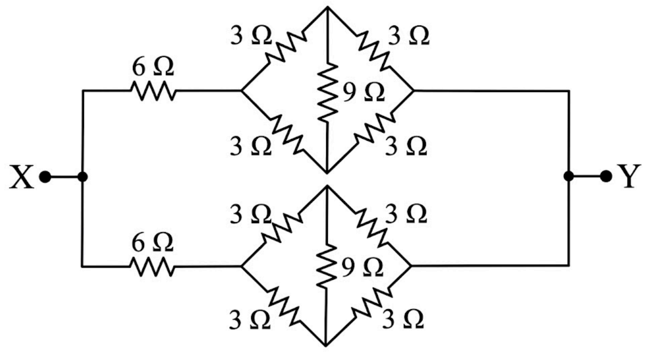
When a metallic surface is illuminated with light of wavelength λ, the stopping potential is V. When the same surface is illuminated with light of wavelength 2λ, the stopping potential becomes $V/4$. The threshold wavelength for this surface is $n\lambda$. The value of n is ____.

Q47.

The displacement of a particle executing simple harmonic motion is $x(t) = A \sin \omega t$. The minimum time taken by the particle to travel from $x = 0$ to $x = A/2$ is t_1, and from $x = A/2$ to $x = A$ is t_2. The value of t_2/t_1 is ____.

Q48.

The equivalent resistance between points X and Y in the given complex circuit is R_{eq}. If R_{eq} is $\frac{n}{2}$ Ω, find the value of n.



Q49.

Two thin prisms are combined with their bases facing in opposite directions to produce dispersion without deviation for the mean ray. The first prism has a refracting angle of 6° and a mean refractive index of 1.5. If the second prism has a mean refractive index of 1.6, its refracting angle is x°. The value of x is _____.

Q50.

A solid cylinder, starting from rest, rolls without slipping down an inclined plane of vertical height 3 m. The linear velocity of its centre at the bottom is $\sqrt{n}$ m/s. The value of n is ____. (Take $g = 10$ m/s^2)

Q51.

An ideal solution is prepared at a constant temperature by mixing equal moles of two volatile liquids, X and Y. The pure vapour pressures of X and Y are 200 torr and 800 torr, respectively. The vapour in equilibrium with this initial solution is completely isolated and condensed to form a new liquid. Calculate the total vapour pressure of this newly condensed liquid at the same temperature.

(1) 500 torr
(2) 320 torr
(3) 680 torr
(4) 1000 torr

Q52.

Identify the CORRECT order of stability for the following carbocations:
(I) $CH_2 = CH - CH_2^+$
(II) $CH_3 - O - CH_2^+$
(III) $CH_2 = CH^+$
(IV) $CH_3 - CH_2^+$

(1) II $>$ I $>$ IV $>$ III
(2) I $>$ II $>$ IV $>$ III
(3) II $>$ IV $>$ I $>$ III
(4) I $>$ IV $>$ II $>$ III

Q53.

Compound A is formed by warming aqueous benzene diazonium chloride. Compound A is then treated with dilute HNO_3 at 298 K to yield a mixture of two isomeric products, B and C. The most suitable method to separate the mixture of B and C is

(1) Simple distillation
(2) Fractional distillation
(3) Steam distillation
(4) Sublimation

Q54.

2.0 moles of an ideal monoatomic gas initially at 300 K undergoes reversible adiabatic compression until its volume is reduced to one-eighth of its initial value. Which of the following options is correct?
(Given: R $= 8.3$ J K^{-1} mol^{-1})

(1) $w = +22.41$ kJ, $\Delta U = +22.41$ kJ, $\Delta H = +37.35$ kJ, $q = 0$
(2) $w = -22.41$ kJ, $\Delta U = -22.41$ kJ, $\Delta H = -37.35$ kJ, $q = 0$
(3) $w = +37.35$ kJ, $\Delta U = +37.35$ kJ, $\Delta H = +22.41$ kJ, $q = 0$
(4) $w = +22.41$ kJ, $\Delta U = 0$, $\Delta H = 0$, $q = -22.41$ kJ

Q55.

Consider the reaction sequence for compound (A) [77.8% C, 7.4% H, vapour density 54], which does not react with sodium metal:
Compound (A) $\xrightarrow{\text{conc. HI, } \Delta}$ Compound (B) + volatile halocarbon
Compound (B) reacts with aqueous bromine to yield a white precipitate.
Identify the INCORRECT statement for the above sequence.

(1) Compound (A) directs incoming electrophiles primarily to the meta position.
(2) Compound (B) gives a characteristic color with neutral $FeCl_3$ solution.
(3) Distilling compound (B) with zinc dust produces an arene.
(4) Compound (A) is synthesized via sodium phenoxide and methyl iodide.

Q56.

Consider a weak polyprotic acid H_3A with $pK_{a1} = 3.0$, $pK_{a2} = 6.0$, and $pK_{a3} = 9.5$. A student needs to prepare a buffer solution with a final pH of 6.0 at 25°C. If 40 mL of a 0.1 M H_3A solution is used, what total volume of 0.1 M NaOH must be added to achieve this?

(1) 20 mL (2) 40 mL (3) 60 mL (4) 80 mL

Q57.

Given below are two statements for the following reaction sequence.

Compound 'X' (Molecular formula C_3H_4) $\xrightarrow{1.\ NaNH_2,\ 2.\ CH_3I}$ Compound 'Y'

$$\begin{cases} \xrightarrow{\quad Na/\text{liq. } NH_3 \quad} \text{Compound 'Z'} \\ \xrightarrow{\quad \text{dil. } H_2SO_4,\ Hg^{2+} \quad} \text{Compound 'Q'} \end{cases}$$

Statement I: Addition of Br_2/CCl_4 to Compound 'Z' produces a racemic mixture.

Statement II: Compound 'Q' yields a yellow precipitate upon reaction with sodium hypoiodite (NaOI).

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is true but Statement II is false
(2) Both Statement I and Statement II are true
(3) Statement I is false but Statement II is true
(4) Both Statement I and Statement II are false

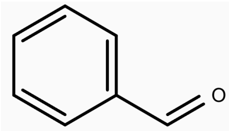
Q58.

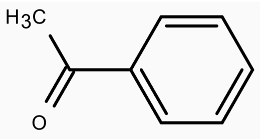
A chemical reaction can proceed via an uncatalyzed pathway or a catalyzed pathway. At a specific temperature T, the rate constant of the catalyzed reaction is e^2 times that of the uncatalyzed reaction. If the activation energy of the uncatalyzed reaction is exactly $4RT$, what is the activation energy of the catalyzed reaction?
(Assume the Arrhenius pre-exponential factor, A, is identical for both pathways).

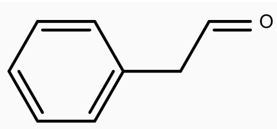
(1) $0.5RT$ (2) $2RT$ (3) $4RT$ (4) $6RT$

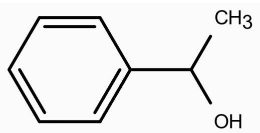
Q59.

Consider the four aromatic compounds (W, X, Y, Z):

(W) 

(X) 

(Y) 

(Z) 

Identify the correct statements.

A. W, X, and Y form a precipitate with 2,4-DNP.

B. Both W and Y reduce Fehling's solution.

C. X and Z give a yellow precipitate with $I_2/NaOH$.

D. W and Y undergo Cannizzaro reaction.

E. W and X can undergo Claisen-Schmidt condensation.

Choose the correct option.

(1) A, B and C only
(2) A, C and E only
(3) C, D and E only
(4) A, D and E only

Q60.

Given below are two statements:

Statement I: The number of non-polar species among XeF_2, ClF_3, BrF_5, and PCl_5 is exactly two.

Statement II: The removal of an electron from an O_2 molecule to form O_2^+ decreases its internuclear distance and transforms it into a diamagnetic species.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement I and Statement II are false
(2) Both Statement I and Statement II are true
(3) Statement I is true but Statement II is false
(4) Statement I is false but Statement II is true

Q61.

Given below are two statements regarding the synthesis of azo dyes:
Statement I: The electrophilic coupling of benzenediazonium chloride with phenol to yield an orange azo dye is optimally conducted in a mildly acidic medium (pH 4-5).
Statement II: In a strongly acidic medium, the coupling reaction of benzenediazonium chloride with aniline is significantly hindered due to the conversion of the activating amino group into a strongly deactivating anilinium ion.
In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is false but Statement II is true
(2) Both Statement I and Statement II are true
(3) Statement I is true but Statement II is false
(4) Both Statement I and Statement II are false

Q62.

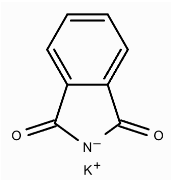
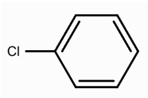
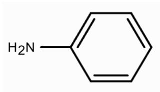
Regarding the general trends in the properties of p-block elements, select the correct statements from the following:
A. The stability of the highest group oxidation state decreases down the group for heavier elements.
B. The first ionization enthalpy of Group 15 elements is generally higher than that of the corresponding Group 16 elements.
C. The metallic character increases significantly as we move down any group in the p-block.
D. The atomic radius strictly and continuously increases from top to bottom in Group 13 elements.
Choose the correct answer from the options given below:

(1) A, B & C only
(2) A & D only
(3) A, B, C & D
(4) B & C only

Q63.

Consider the following reaction sequences giving a major product. Identify the correct reaction.

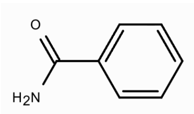
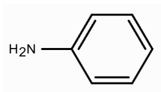
(1)

 $+$  $\xrightarrow[\text{(ii) } H_3O^+]{\text{(i) Heat}}$ 

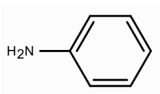
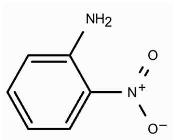
(2)

$$CH_3CH_2-NH_2 \xrightarrow{CHCl_3/KOH,\Delta} \text{Intermediate} \xrightarrow{Sn/HCl,\Delta} CH_3CH_2CH_2-NH_2$$

(3)

 $+Br_2+4KOH(alc) \xrightarrow{\Delta}$  $+2KBr+K_2CO_3+2H_2O$

(4)

 $\xrightarrow{HNO_3/H_2SO_4,288K}$  (Major Product)

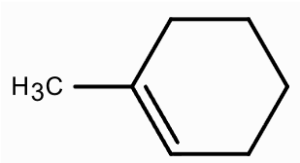
Q64.

The de Broglie wavelengths of electrons in various orbits of an H atom are considered. Identify the set of wavelengths belonging to the first three excited states.
(a_0 = Bohr radius)

(1) $4\pi a_0, 6\pi a_0, 8\pi a_0$
(2) $2\pi a_0, 4\pi a_0, 6\pi a_0$
(3) $8\pi a_0, 18\pi a_0, 32\pi a_0$
(4) $4\pi a_0, 9\pi a_0, 16\pi a_0$

Q65.

Consider the following reaction sequence:



$\xrightarrow{\text{(i) } B_2H_6/\text{THF, (ii) } H_2O_2/OH^-}$ Product 'P'

Consider the following statements regarding the above reaction:

A. The reaction involves the formation of a classical carbocation intermediate.

B. The overall process results in the syn-addition of H and OH across the double bond.

C. The major product 'P' is capable of exhibiting optical isomerism.

D. The regioselectivity of this reaction is governed by the formation of a highly stable free radical.

E. Oxymercuration-demercuration of the same reactant yields a different major constitutional isomer without skeletal rearrangement.

Identify the correct statements. Choose the correct answer from the options given below:

(1) B, C & E Only
(2) A, B & D Only
(3) C, D & E Only
(4) A, C & E Only

Q66.

In the given tetrapeptide, find out the amino acid (Z) capable of forming disulfide cross-links and identify the correct sequence present:

$H_2N\text{-}CH(CH(CH_3)_2)\text{-}CO\text{-}NH\text{-}CH(CH_2SH)\text{-}CO\text{-}NH\text{-}CH(CH_2C_6H_5)\text{-}CO\text{-}NH\text{-}CH(CH_2CH_2COOH)\text{-}COOH$

Choose the correct answer from the options given below:

(1) Z = Cysteine; Sequence: Val - Cys - Phe - Glu
(2) Z = Methionine; Sequence: Leu - Cys - Tyr - Asp
(3) Z = Cysteine; Sequence: Val - Cys - Tyr - Glu
(4) Z = Valine; Sequence: Val - Cys - Phe - Glu

Q67.

Given below are two statements regarding transition and inner transition elements:
Statement I: The $E^{\circ}(M^{3+}/M^{2+})$ value for Manganese (Mn) is highly positive compared to that of Chromium (Cr) and Iron (Fe) because the half-filled d^5 configuration of the Mn^{2+} ion provides exceptional thermodynamic stability against further oxidation.
Statement II: The steady decrease in the ionic radii across the lanthanoid series results in a gradual increase in the covalent character of their hydroxides, thereby making $Lu(OH)_3$ significantly more basic than $La(OH)_3$.
In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is true but Statement II is false
(2) Statement I is false but Statement II is true
(3) Both Statement I and Statement II are false
(4) Both Statement I and Statement II are true

Q68.

An atomic orbital of a hydrogen atom possesses a total of 3 nodes and a single angular nodal plane. What is the orbital angular momentum of an electron in this orbital, and how many regions of maximum electron probability (peaks) exist in its radial distribution curve?

(1) $\sqrt{2}\frac{h}{2\pi}$ and 3
(2) 0 and 4
(3) $\sqrt{6}\frac{h}{2\pi}$ and 2
(4) $\sqrt{2}\frac{h}{2\pi}$ and 2

Q69.

The correct statement regarding the geometry of the following complexes is:

(1) $[PtCl_4]^{2-}$ and $[NiCl_4]^{2-}$ are square planar and $[Zn(NH_3)_4]^{2+}$ is tetrahedral.
(2) $[PtCl_4]^{2-}$ and $[Zn(NH_3)_4]^{2+}$ are tetrahedral and $[NiCl_4]^{2-}$ is square planar.
(3) $[PtCl_4]^{2-}$ is square planar and $[NiCl_4]^{2-}$ and $[Zn(NH_3)_4]^{2+}$ are tetrahedral.
(4) $[NiCl_4]^{2-}$ is square planar and $[PtCl_4]^{2-}$ and $[Zn(NH_3)_4]^{2+}$ are tetrahedral.

Q70.

In period 3 of the periodic table, the elements with highest first ionization energy and highest electron affinity are respectively:

(1) Ar & Cl
(2) Ne & F
(3) Cl & Ar
(4) Ar & S

Q71.

X is the number of unpaired electrons present in $[Fe(CN)_6]^{3-}$.
Y is the primary valency of the central metal in $[Co(NH_3)_5Cl]Cl_2$.
Z is the number of five-membered chelate rings in $[M(EDTA)]^{2-}$.
The value of $X+Y+Z$ is ____.

Q72.

A 500 mL solution containing a mixture of 0.4 M $FeSO_4$ and 0.2 M FeC_2O_4 is prepared in an acidic medium. This entire mixture is completely oxidized by titrating it against a standard 0.05 M $KMnO_4$ solution till a faint permanent pink color appears. The volume (in L) of $KMnO_4$ consumed is ____.

Q73.

Consider the following redox reaction taking place in acidic medium:
$Cr_2O_7^{2-}(aq) + Fe^{2+}(aq) \rightarrow Cr^{3+}(aq) + Fe^{3+}(aq)$
If the Nernst equation for the balanced reaction (with simplest whole-number coefficients) is
$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \dfrac{RT}{nF}\ln Q$
and the denominator of Q contains the term $[H^+]^y$,
Then the value of $(y-n)$ is ____.

Q74.

0.4 g of an organic compound (Z) gave 0.932 g of $BaSO_4$ precipitate in Carius estimation for sulfur. In Kjeldahl's estimation, the ammonia evolved from 0.5 g of (Z) was completely neutralized by 25 mL of 0.1 M H_2SO_4. Further, 0.5 g of (Z) gave 0.88 g of CO_2 gas on complete combustion. The percentage of hydrogen in compound (Z) is ____ %.
[Given: Molar mass in g mol^{-1} H: 1, C: 12, N: 14, S: 32, O: 16, Ba: 137; Compound (Z) contains only C, H, N, and S]

Q75.

Consider the dissolution equilibrium of the sparingly soluble salt $AB_2 \rightleftharpoons A^{2+}(aq) + 2B^-(aq)$.
If the molar solubility of the salt AB_2 in pure water is 10^{-3} M, and its solubility product K_{sp} is expressed as $X \times 10^{-9}$, then the value of X is ____.

ANSWERS KEY

1. (1)	**2.** (3)	**3.** (1)	**4.** (1)	**5.** (3)	**6.** (1)	**7.** (2)	**8.** (2)
9. (1)	**10.** (1)	**11.** (3)	**12.** (3)	**13.** (2)	**14.** (2)	**15.** (3)	**16.** (2)
17. (1)	**18.** (1)	**19.** (1)	**20.** (2)	**21.** 55.0	**22.** 1.0	**23.** 90.0	**24.** 12.0
25. 120.0	**26.** (1)	**27.** (2)	**28.** (1)	**29.** (3)	**30.** (3)	**31.** (2)	**32.** (3)
33. (2)	**34.** (2)	**35.** (2)	**36.** (3)	**37.** (2)	**38.** (3)	**39.** (3)	**40.** (3)
41. (3)	**42.** (3)	**43.** (1)	**44.** (1)	**45.** (3)	**46.** 3.0	**47.** 2.0	**48.** 9.0
49. 5.0	**50.** 40.0	**51.** (3)	**52.** (1)	**53.** (3)	**54.** (1)	**55.** (1)	**56.** (3)
57. (3)	**58.** (2)	**59.** (2)	**60.** (3)	**61.** (1)	**62.** (1)	**63.** (3)	**64.** (1)
65. (1)	**66.** (1)	**67.** (1)	**68.** (1)	**69.** (3)	**70.** (1)	**71.** 9.0	**72.** 2.0
73. 8.0	**74.** 6.0	**75.** 4.0					